

Topic to discuss

- Gauss Backward central difference Interpolation
- Gauss Backward Table
- Gauss Backward Formula
- Numerical Problem
- Homework Problem

Gauss Backward Difference Interpolation

A central-difference interpolation method for equally spaced data used when the target x lies just behind a chosen central tabular point. It expands $y(x)$ in the same $u = \frac{x-x_0}{h}$ but uses backward central differences, making it well suited for interior points slightly below x_0 while retaining the accuracy benefits of central schemas.

When to use :

- Use it when x lies between x_{-1} and x_0
i.e, $-1 < u < 0$

Gauss Backward Table

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_{-2}	y_{-2}	$\nabla y_{-2} = y_{-1} - y_{-2}$	$\nabla^2 y_{-2} = \nabla y_{-1} - \nabla y_{-2}$	$\nabla^3 y_{-2} = \nabla^2 y_{-1} - \nabla^2 y_{-2}$	$\nabla^4 y_{-2} = \nabla^3 y_{-1} - \nabla^3 y_{-2}$
x_{-1}	y_{-1}	$\nabla y_{-1} = y_0 - y_{-1}$	$\nabla^2 y_{-1} = \nabla y_0 - \nabla y_{-1}$	$\nabla^3 y_{-1} = \nabla^2 y_0 - \nabla^2 y_{-1}$	
x_0	y_0	$\nabla y_0 = y_1 - y_0$	$\nabla^2 y_0 = \nabla y_1 - \nabla y_0$		
x_1	y_1	$\nabla y_1 = y_2 - y_1$			
x_2	y_2				

Gauss Backward Central difference Interpolation Formula

The Formula is :

$$y(x) = y_0 + u \nabla y_{-1} + \frac{u(u+1)}{2!} \nabla^2 y_{-1} + \frac{(u-1)u(u+1)}{3!} \nabla^3 y_{-2} + \frac{(u-1)u(u+1)(u+2)}{4!} \nabla^4 y_{-2} + \dots$$

$$\text{where, } u = \frac{x - x_0}{h}$$

and h is constant difference
between successive values of x .

Also,

$-1 < u < 0 \Rightarrow$ for best
approximate value.

Numerical Problem

Q: Using the table, find $\sqrt{9.6}$ by Gauss backward central Interpolation.

x	8	9	10	11	12
$\sqrt{x}$	2.828427	3.000000	3.162278	3.316625	3.464102

Solution: step length: $h = 9 - 8 = 1$

Choose central value: $x_0 = 10$

Interpolation point: $x = 9.6$

$$u = \frac{x - x_0}{h} = \frac{9.6 - 10}{1} = -0.4$$

Gauss backward central difference table,

	x	$y = f(x) = \sqrt{x}$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-2}	8	2.828427	0.171573	-0.009295		
x_{-1}	9	3.000000	0.162278	-0.007931	0.001364	-0.000303
x_0	10	3.162278	0.154347	-0.00687	0.001061	
x_1	11	3.316625	0.147477			
x_2	12	3.464102				

Gauss Backward central difference Interpolation,

$$y(x) = y_0 + u \nabla y_{-1} + \frac{u(u+1)}{2!} \nabla^2 y_{-1} + \frac{(u-1)u(u+1)}{3!} \nabla^3 y_{-2} + \frac{(u-1)u(u+1)(u+2)}{4!} \nabla^4 y_{-2}$$

$$= 3.162278 + (-0.4) \times 0.162278 + \frac{-0.4(-0.4+1)}{2!} \times -0.007931 + \frac{(-0.4-1)(-0.4)(-0.4+1)}{3!} \times 0.001364$$
$$+ \frac{(-0.4-1)(-0.4)(-0.4+1)(-0.4+2)}{4!} \times -0.000303$$

$$= 3.162278 - 0.0649112 + 0.000951720 + 0.000076384 - 0.000006787$$

$$= 3.098388117$$

Ans.

Homework Problem

Q:- Using the following table of e^x , find $e^{0.26}$

x	0.0	0.1	0.2	0.3	0.4
e^x	1.000	1.105	1.221	1.350	1.492

Solution : Step length, $h = (0.1 - 0.0) = 0.1$

Choose central value: $x_0 = 0.3$

Interpolation point, $x = 0.26$

$$\therefore u = \frac{x - x_0}{h} = \frac{0.26 - 0.3}{0.1}$$

$$u = -0.4$$

Gauss backward central difference table,

	x	$y = f(x) = e^x$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-3}	0.0	1.000	0.105	0.011	0.002	
x_{-2}	0.1	1.105	0.116	0.013	0.000	-0.002
x_{-1}	0.2	1.221	0.129	0.013		
x_0	0.3	1.350	0.142			
x_1	0.4	1.492				

Gauss Backward central difference Interpolation,

$$y(x) = y_0 + u \nabla y_{-1} + \frac{u(u+1)}{2!} \nabla^2 y_{-1} + \frac{(u-1)u(u+1)}{3!} \nabla^3 y_{-2} + \frac{(u-1)u(u+1)(u+2)}{4!} \nabla^4 y_{-2}$$

$$= 1.350 + (-0.4)(0.129) + \frac{(-0.4)(-0.4+1)}{2!} \times 0.013$$

$$= 1.350 - 0.0516 - 0.00156$$

$$= 1.29684$$

Ans.

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